# RetinaScreen: AI-Assisted Clinical Screening Report

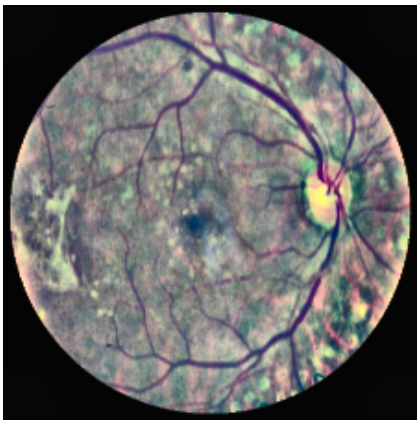
Diabetic Retinopathy Automated Triaging & Referral Decision Support

| | |
|---|---|
| **Report ID:** RS-1793825700 | **Date/Time:** 2026-11-05 02:25:00 |
| **Image Quality:** Passed: Adequate Diagnostic Quality (Passed T2 Quality Gate) | **Blur Score:** 536.0 |

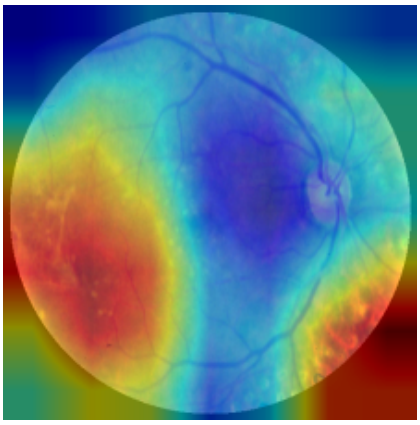
## 1. Diagnostic Finding & Referral Recommendation

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|---|---|
| **Predicted ICDR Severity Grade:** | **Grade 4: Proliferative DR** |
| **Deep Learning Confidence:** | 37.1% (Low Confidence Borderline Case) |
| **Clinical Triage Action:** | **Emergency Vitreoretinal Consultation** |
| **Target Referral Window:** | Within 24 - 48 hours |

## 2. Imaging Diagnostics & Explainable AI Attention



**CLAHE Enhanced Fundus**



**Grad-CAM Lesion Heatmap Overlay**

## 3. Multimodal Lifestyle & Retinopathy Risk Fusion *(SIMULATED)*

| | |
|---|---|
| **Multimodal Risk Score (0 - 100):** | **70.7 / 100** |
| **Assigned Behavioral Phenotype:** | Profile 3: Younger High-Calorie At-Risk |
| **Clinical Interpretation:** | Integrated linear opinion pool weighting fundus CNN features (70%) with systemic lifestyle cofactors (30%). |

***ACADEMIC RESEARCH & CLINICAL DISCLAIMER:*** *This screening report was generated programmatically by RetinaScreen as part of an educational advanced AI research project. It does NOT constitute medical advice, definitive clinical diagnosis, or physician prescription. Automated AI triage serves strictly as an auxiliary screening decision-support aid. All visual findings and referral decisions must be independently reviewed and confirmed by a licensed, board-certified ophthalmologist or optometrist.*